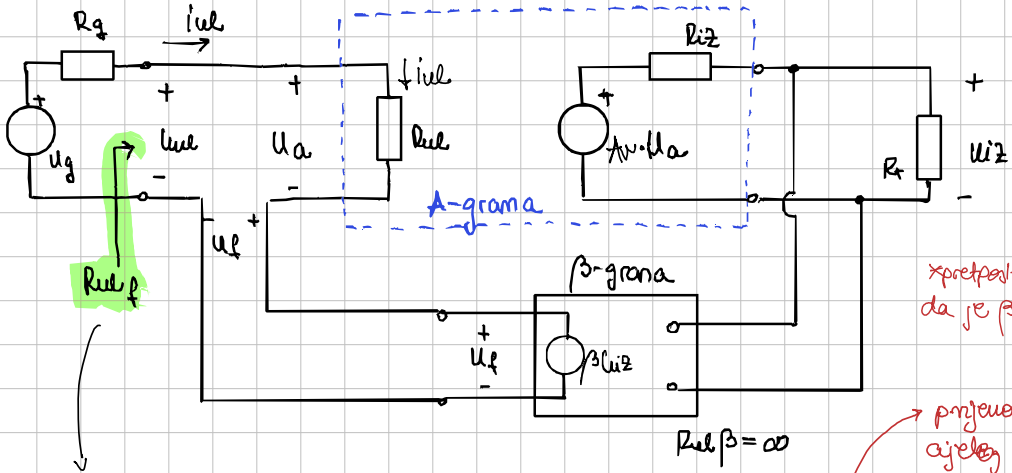
Kako povratna veza utječe na ulazne / izlazne otpore

NAPONSKA – SERIJSKA PV – Ulazni otpor

izlazna: napon (mjerimo (V) → β-grani spajamo PARALELNO pretvarač)

ulazna: napon (zbrajanje napona (Σ) → SERIJA)

→ osnovno pojačalo = NAPONSKO (pojačalo u A-grani)



*pretpostavljamo da je β-grana idealna

$$R_{ul f} = \frac{u_{ul}}{i_{ul}} = \frac{u_a + u_f}{\frac{u_a}{R_{ul}}} = R_{ul}\left(\frac{u_a + u_f}{u_a}\right) = R_{ul}\left(1 + \frac{u_f}{u_a}\right)$$

→ prijenosna fja cijelog pojačala između te dvije točke
↳ A-grana i cijela β-grana

$\Rightarrow \beta \cdot A_v$

$A_v = \frac{u_{iz}}{u_a}$ (vrijedi za opterećen izlaz), $A_v = ?$

A_v vrijedi za odspojeni R_T ($R_T = \infty$) → $A_v = A_v\big|_{R_T=\infty}$

$$R_{ul f} = R_{ul}\left(1 + \frac{u_f}{u_{iz}} \cdot A_v\right) \Rightarrow \boxed{R_{ul f} = R_{ul}(1 + \beta A_v)}$$

serijska NPV povećava $R_{ul f}$

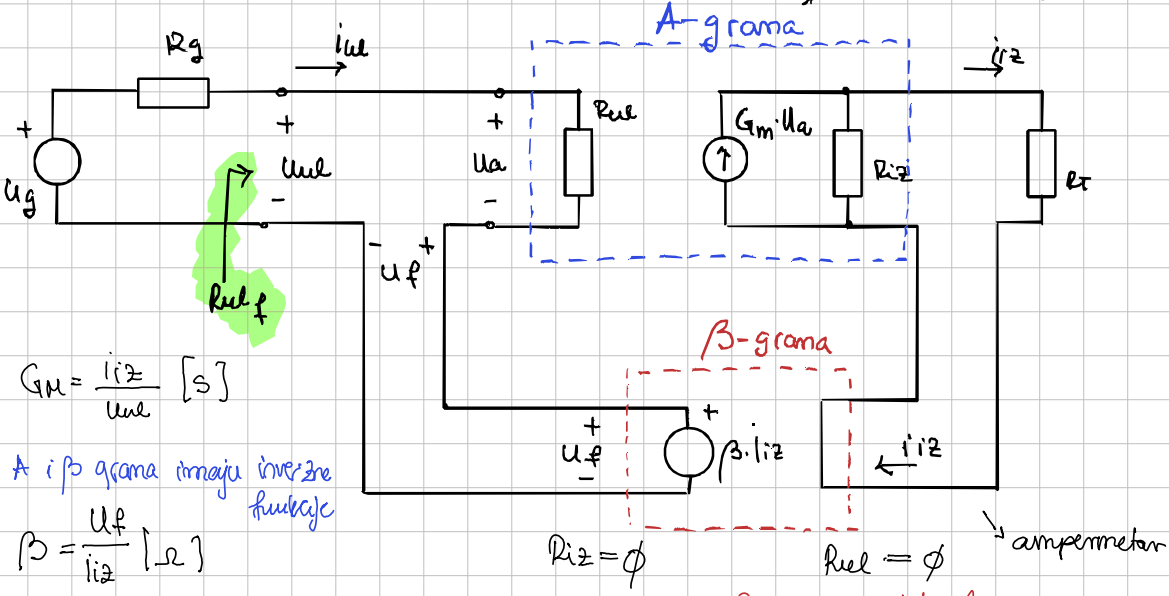
STRUJNA - SERIJSKA PV

Ulazni otpor

ulaz: napon (zbrajam u SERIJI)

izlaz: struja (mjerimo AMPERMETROM)

$\frac{i_{ul}}{u_{ul}}$ (GM) → strmina ⇒ STRMINSKO POJAČALO ⇓ A-grana



$G_M = \frac{i_{iz}}{u_{ul}}$ [S]

A i β grana imaju inverzne funkcije

$\beta = \frac{u_f}{i_{iz}}$ [Ω]

$R_{iz} = \phi$ $R_{ul} = \phi$

* β-grana → idealna (ne opterećuje pojačalo u A-grani)

$$R_{ulf} = \frac{u_{ul}}{i_{ul}} = \frac{u_{ul}}{\frac{u_a}{R_{ul}}} = R_{ul} \cdot \frac{u_{ul}}{u_a}$$

→ ulaznu struju zapisujemo pomoću struje koja ulazi u A-granu

$$R_{ulf} = R_{ul}\left(\frac{u_a + u_f}{u_a}\right) = R_{ul}\left(1 + \frac{u_f}{u_a}\right) \Longrightarrow \boxed{R_{ulf} = R_{ul}(1 + \beta G_M)}$$

* $G_m \neq G_M \longrightarrow G_m = G_M \big|_{R_T = \phi}$ (maks moguće strminsko pojačanje)

→ Ulazni otpori za serijski tip veze RASTU

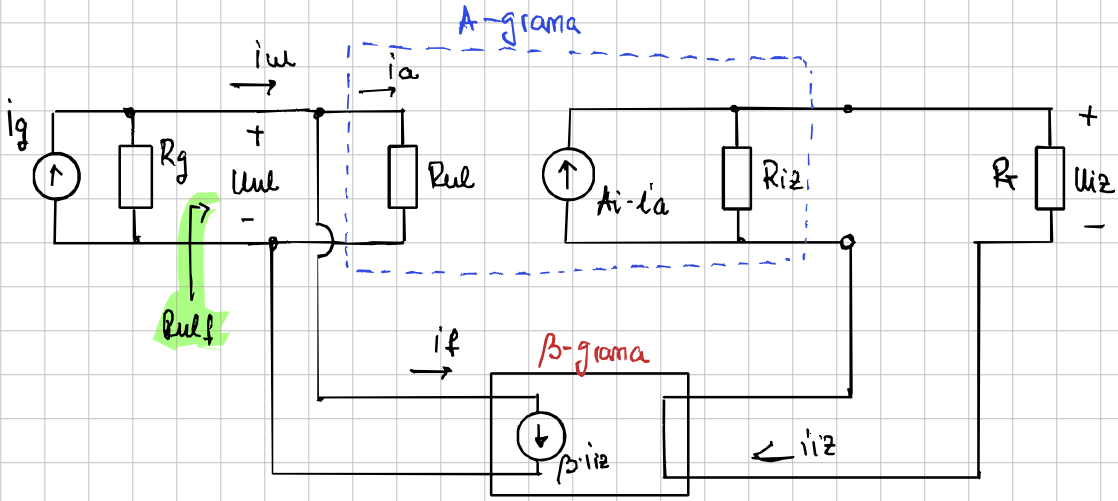
STRUJNA - PARALELNA

Ulazni otpor

Izlaz: struja (AMPERMETAR → β-grana = SERIJA)

Ulaz: struja (Zbrajanje PARALELOM)

} strujno pojačalo



$$A_I = \frac{i_{iz}}{i_a} \Rightarrow A_i = A_I\big|_{R_t=0}$$

$$\frac{1}{R_{ulf}} = \frac{i_{ul}}{u_{ul}} = \frac{i_a + i_f}{i_a \cdot R_{ul}} = \frac{1}{R_{ul}}\left(1 + \frac{i_f}{i_a}\right) = \frac{1}{R_{ul}}\left(1 + \underbrace{\frac{i_f}{i_{iz}}}_{\beta} \cdot \underbrace{\frac{i_{iz}}{i_a}}_{A_I}\right)$$

$$\Rightarrow \boxed{R_{ulf} = \frac{R_{ul}}{1 + \beta A_I}}$$

Paralelna veza → smanjuje R_{ulf} ↓↓

NAPONSKA - PARALELNA

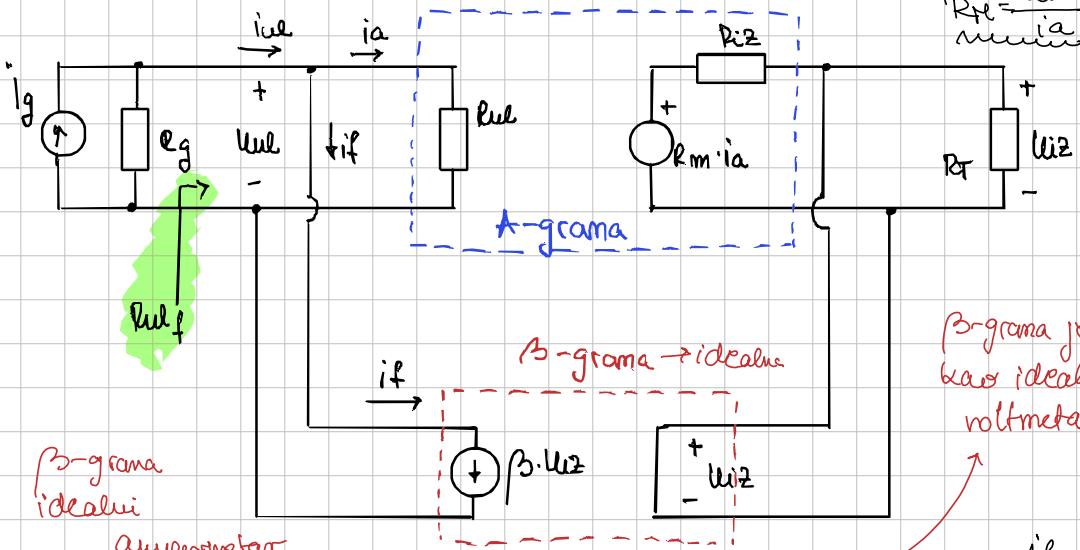
· Ulazni otpor

izlaz: napon - mjerimo VOLTMETROM (spajamo PARALELNO)

ulaz: struja - zbrajamo u PARALELI

OTPORNO POJAČALO

$R_M = \frac{u_{iz}}{i_a}$



B-grama je kao idealni voltmetar

B-grama idealni ampermetar

$R_{iz} = \infty$

$R_{ul} = \infty$

$\beta = \frac{i_f}{u_{iz}}$

$R_M = \frac{u_{iz}}{i_a}$, $R_m \to$ uz opterećeno pojačalo $\Rightarrow R_m = R_M \big|_{R_T = \infty}$

$$\frac{1}{R_{ulf}} = \frac{i_{ul}}{u_{ul}} = \frac{i_f + i_a}{i_a \cdot R_{ul}} = \frac{1}{R_{ul}}\left(1 + \underbrace{\frac{i_f}{i_a}}_{\beta R_M}\right) = \frac{1}{R_{ul}}\left(1 + \underbrace{\frac{i_f}{u_{iz}}}_{\beta} \cdot \underbrace{\frac{u_{iz}}{i_a}}_{R_M}\right)$$

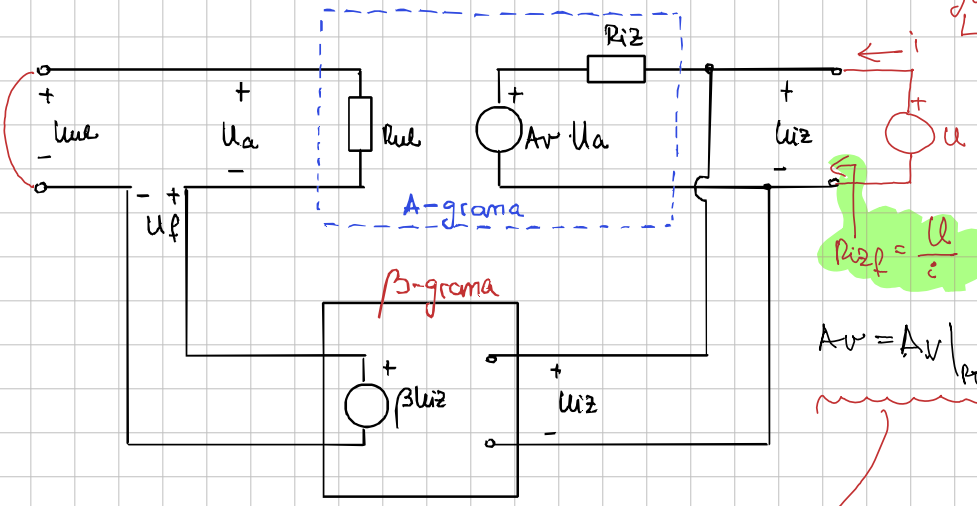
$$\Rightarrow \boxed{R_{ulf} = \frac{R_{ul}}{(1 + \beta \cdot R_M)}}$$

$R_{ulf} \downarrow\downarrow$

NAPONSKA - SERIJSKA PV

Izlazni otpor • K.S. ulaz

• gasimo pobuđač
↳ naponski izvor



$$R_{izf} = \frac{u}{i}$$

$$A_v = A_v\big|_{R_L=\infty}$$

$R_{izf\beta} = \phi$ $\quad R_{ul\beta} = \infty$

idealna β-grana

malo A_v!

malo v!

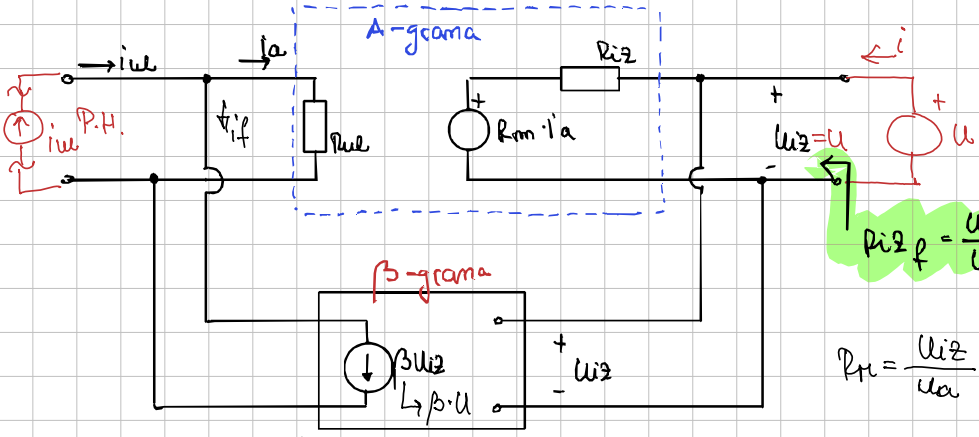
$$i = \frac{u - A_v \cdot U_a}{R_{iz}} = \frac{u - A_v(-\beta \cdot u)}{R_{iz}} = \frac{u(1 + A_v\beta)}{R_{iz}} \Rightarrow \boxed{R_{izf} = \frac{R_{iz}}{1+\beta A_v}}$$

$U_a = -U_f = -\beta u$

naponska PV smanjuje R_{izf}

NAPONSKA - PARALELNA

Izlazni otpor



$$R_{izf} = \frac{u}{i}$$

$$R_M = \frac{U_{iz}}{U_a}$$

$$R_m = R_M\big|_{R_L=\infty}$$

$R_{izf\beta} = \infty$ $\quad R_{ul\beta} = \infty$

idealna β-grana

$$i = \frac{u - R_m \cdot i_a}{R_{iz}}$$

$$i_{ul} = i_a + i_f$$

$$\phi \rightarrow i_a = -i_f = -\beta \cdot u$$

$$i = \frac{u - R_m(-\beta u)}{R_{iz}}$$

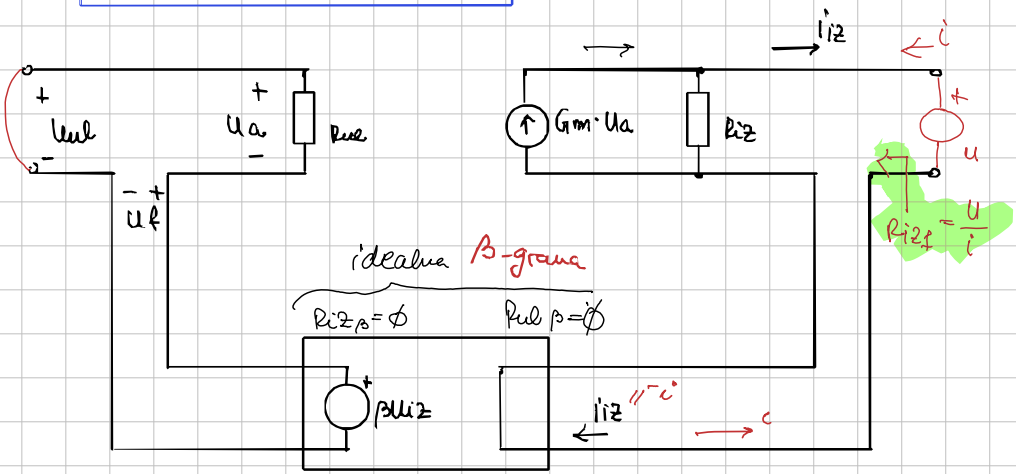
$$i = \frac{u}{R_{iz}}(1 + R_m\beta)$$

$$\Rightarrow \boxed{R_{izf} = \frac{R_{iz}}{1+\beta \cdot R_m}}$$

naponska NPV smanjuje R_{izf} ↓

STRUJNA - SERIJSKA

Izlazni otpor



$$u = (i + G_m \cdot u_a) R_{iz} = (i + G_m \cdot \beta i) R_{iz} = i(1 + \beta G_m) R_{iz}$$

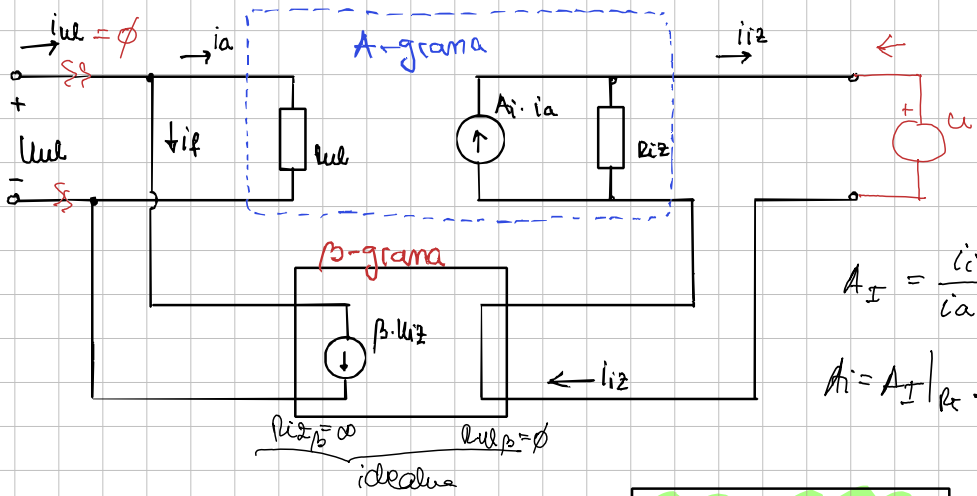
$$u_a = -u_f = -(-\beta i) = \underline{\underline{\beta i}}$$

$$\Rightarrow \boxed{R_{izf} = (1 + \beta G_m) R_{iz}}$$

Strujna NPV povećava $\underline{\underline{R_{izf}}}$ ↑

STRUJNA - PARALELNA PV

Izlazni otpor



$$A_I = \frac{i_{iz}}{i_a}$$

$$A_i = A_I \big|_{R_t = \emptyset}$$

$$u = (i + A_i \cdot i_a) R_{iz} = (i + A_i \cdot \beta i) R_{iz} \rightarrow$$

$$i_a = -i_f = -\beta i$$

$$\boxed{R_{izf} = R_{iz}(1 + \beta A_i)}$$

Strujna PV povećava R_{izf} ↑